

Design and Realization of a 4 KB (1024 × 32) SRAM Using OpenRAM and SKY130 PDK: An Industry-Standard Realization Roadmap

Pavan Kumar K A

Abstract— This work presents an industry-standard roadmap for the design and realization of a 4 KB (1024 × 32) SRAM using the OpenRAM compiler and the SKY130 PDK. The focus is to understand the end-to-end SRAM design flow, including architecture, 6T bitcell operation, peripheral circuits, OpenRAM flow, characterization, and verification. Industry metrics such as SNM, write margin, read disturb, and variation analysis are emphasized. The roadmap integrates industry practices for debugging and AI-assisted productivity. RISC-V integration and FPGA fabric exploration are reserved for future work.

Index Terms— SRAM, OpenRAM, SKY130, 6T bitcell, SNM, Write Margin, Characterization, Verification, Variation.

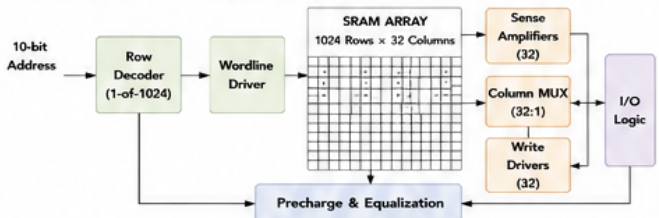
1. REQUIREMENT ANALYSIS

- Capacity : $1024 \times 32 = 32,768$ bits = 4 KB (32 Kbit)
- Address Width : $\log_2(1024) = 10$ bits
- Data Width : 32 bits
- Technology : SKY130 (1.8 V)
- Target Access Time : < 2.5 ns
- Application : On-chip memory (general purpose)

Why SRAM?

Memory Type	Speed	Density	Refresh	Use Case
Register File	Very High	Low	No	CPU Core
SRAM	High	Medium	No	Cache, Buffer
DRAM	Medium	High	Yes	Main Memory
Flash	Low	High	No	Non-volatile

2. SRAM ARCHITECTURE (1024 × 32)

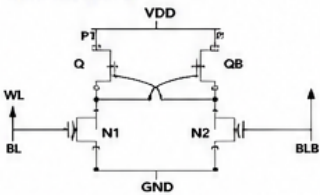


Block Functions

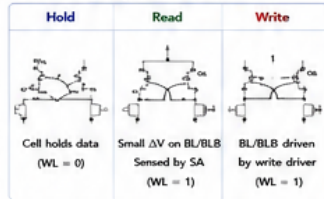
- Row Decoder** : Selects one wordline from 1024.
- Wordline Driver** : Drives the selected wordline with sufficient strength.
- SRAM Array** : Stores data in 6T bitcells.
- Sense Amplifier** : Detects small differential on bitlines.
- Column MUX** : Selects one of 32 column outputs.
- Write Drivers** : Drive data during write operation.
- Precharge/Equalization** : Precharges and equalizes bitlines.

3. 6T SRAM BITCELL & KEY METRICS

3.1 6T SRAM Bitcell



3.2 Operating Modes



3.3 Key Metrics (Industry)

- Hold SNM (Static Noise Margin)
- Read SNM
- Write SNM
- Butterfly Curve
- Cell Ratio ($\beta = W_{PDD}/W_{ACC}$)
- Pull-Up Ratio ($\alpha = W_{PDD}/W_{PDD}$)
- Write Margin
- Read Disturb / Write Disturb
- Half-Select Disturb
- Leakage (Standby)
- Data Retention (Low VDD)

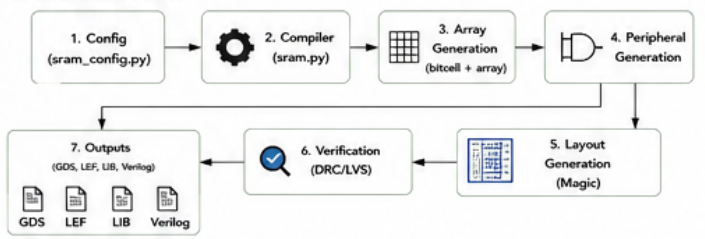
4. PERIPHERAL CIRCUITS OVERVIEW

Circuit	Function	Inputs	Outputs	Key Considerations
Precharge & Equalization	Precharge and equalize bitlines	PRE	BL, BLB	Precharge time, leakage
Row Decoder	Decode 10-bit address to 1-of-1024 WL	Addr[9:0]	WL[1023:0]	Delay, area, glitch
Wordline Driver	Drive wordline with sufficient strength	WL_in	WL_out	Fanout, rise/fall time
Sense Amplifier (Latch Type)	Sense small ΔV on bitlines	BL, BLB	Data_out	Offset, speed, noise
Write Driver	Drive bitlines during write	Data_in	BL, BLB	Write strength, failure
Column MUX (32:1)	Select one of 32 column outputs	Sel[4:0]	Data_out	Area vs speed trade-off

12. AI-ASSISTED WORKFLOW

ChatGPT	GitHub Copilot	Python	TCL / Shell
- Concept Explanation - Architecture Clarification - Debug Guidance - Report Generation	- Code Snippets - Script Generation - Error Analysis - Automation	- Data Parsing - Plotting & Analysis - Report Generation - Automation	- Tool Flow Automation - Batch Runs - Log Parsing - Report Generation

5. OPENRAM FLOW



5.1 Important OpenRAM Files

File	Purpose
sram.py	Top-level SRAM generator and flow control
sram_config.py	User configuration (size, tech, options)
bitcell.py	Bitcell description
tech.py	Technology abstraction (layers, rules, models)
compiler.py	Compiler engine
characterizer/	Timing & power characterization
globals.py	Global parameters
analysis/	Post-layout analysis

5.2 Key Configuration Parameters

Parameter	Description	Example
word_size	Number of bits per word	32
num_words	Number of words	1024
tech_name	Technology (e.g., sky130A)	sky130A
supply_voltage	Supply voltage (V)	1.8
num_banks	Number of banks	1
write_size	Write port size	32

5.3 Process Corners (SKY130)

Corner	Description	Use
TT	Typical process, typical V, 25°C	Nominal
SS	Slow-Slow	Worst delay
FF	Fast-Fast	Best delay
SF	Slow-Fast	Read failures
FS	Fast-Slow	Write failures

6. SKY130 TECHNOLOGY OVERVIEW

6.1 Device Models

- NMOS & PMOS (BSIM4 Models)
- Accurate transistor behavior

6.2 Layer Stack (Simplified)



6.3 Technology Files

- SPICE Models
- LEF (Physical Abstract)
- GDS (Layout)
- LIB (Timing/Power)
- Tech LEF, DRC Rules, L VS Rules

6.4 Process Views

- Multiple PVT corners
- DRC clean layout
- LVS clean netlist
- Parasitic extraction (PEX)

7. GENERATED OUTPUTS

GDSII Layout	SPICE Netlist	LEF File	LIB File	Verilog Model	Reports
Physical layout for fabrication	Circuit-level netlist for simulation	Physical abstract for P&R	Timing, power, area information	RTL model for simulation	Area, timing, power, DRC, LVS, etc.

8. VERIFICATION FLOW

8.1 Functional Verification	8.2 Physical Verification	8.3 Electrical Verification
- Read Operation - Write Operation - Hold Operation - March Tests	- DRC (Design Rule Check) - LVS (Layout vs Schematic) - PEX (Parasitic Extraction)	- Timing (Setup, Hold, Access) - Power (Read, Write, Leakage) - Noise / IR Drop - Signal Integrity

9. CHARACTERIZATION

9.1 Timing	9.2 Power	9.3 Outputs
- Access Time (Read/Write) - Setup Time - Hold Time - Cycle Time	- Read Dynamic Power - Write Dynamic Power - Leakage Power - Standby Power	- LIB (Liberty) - LEF (Physical) - GDS (Layout) - SPICE / Verilog

Advanced Models: NLDL | CCS | LVF

10. VARIATION ANALYSIS

- Process Variation (Vth, LER)
- Voltage Variation
- Temperature Variation
- Monte Carlo Simulation
- Mismatch (Vth, W/L)
- Yield Estimation

11. DEBUG & FAILURE ANALYSIS

- OpenRAM Issues (config, version, modules)
- PDK / Setup Issues (paths, models)
- DRC Violations (design rule errors)
- LVS Mismatches (connectivity, pins)
- SPICE Issues (convergence, includes)
- Timing Failures (decoder, bitline, SA delay)

13. CONCLUSION

This roadmap provides a complete industry-standard approach to design and realize a 4 KB (1024 × 32) SRAM using OpenRAM and SKY130. It covers architecture understanding, circuit operation, compiler flow, characterization, verification, variation analysis, and debugging with industry best practices and AI-assisted productivity.

14. FUTURE WORK

- Integrate SRAM into a RISC-V based SoC
- Explore FPOA fabric design using SRAM (LUT, routing, configuration memory)
- Investigate advanced memories (e.g., SRAM with ECC, multi-port SRAM)
- Perform silicon measurement after fabrication
- Extend flow to advanced nodes and tools